

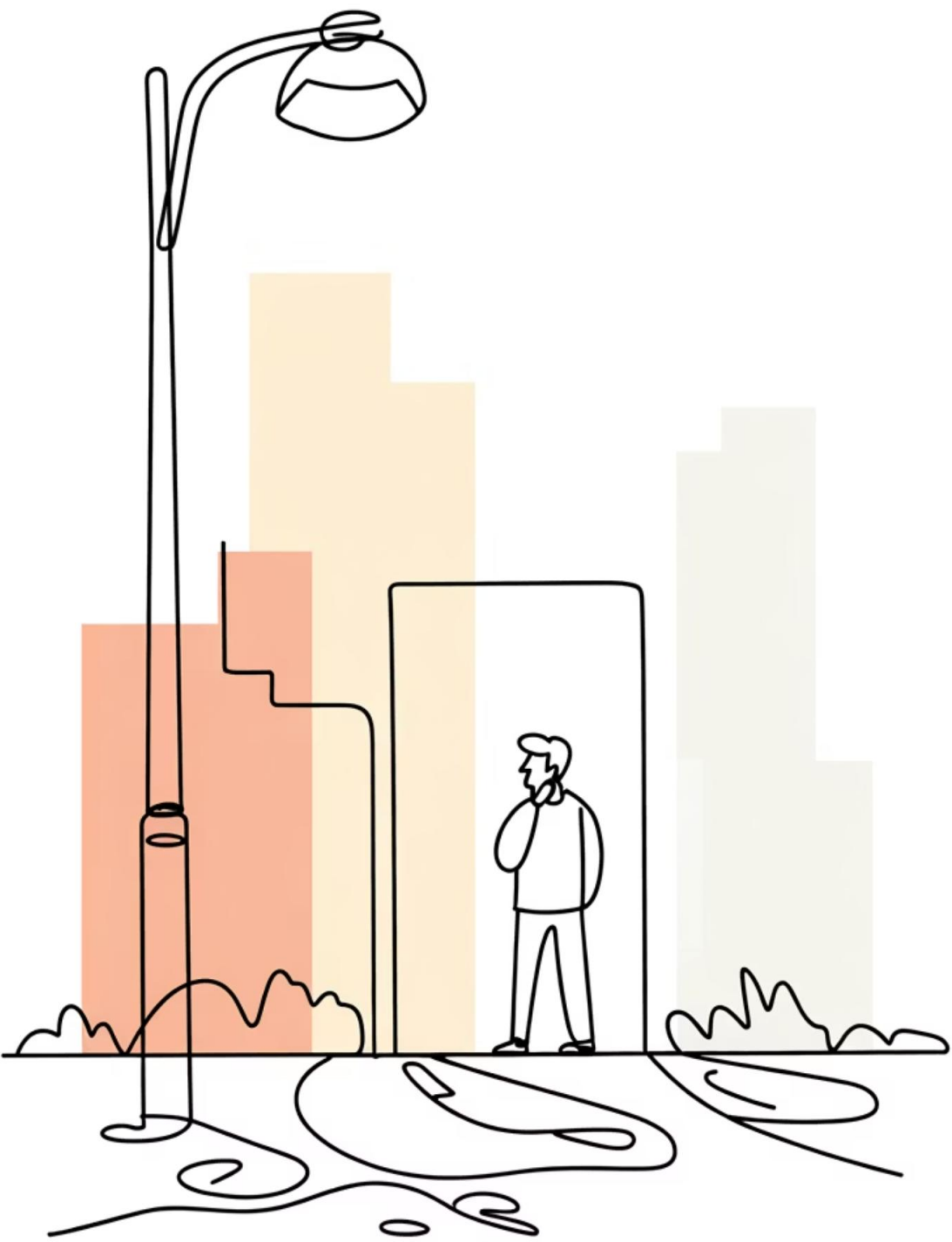


CivicConnect – AI-Powered Citizen Reporting App

Smart Governance Through Citizen Participation

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SIH ID: SIH25031



The Challenge: Unreported and Unresolved Civic Issues

Invisible Problems

Potholes, broken streetlights, or sanitation issues often go unseen by city authorities, leading to prolonged neglect.

Cumbersome Reporting

Existing reporting systems are typically slow, hard to navigate, or demand lengthy, non-transparent complaint procedures.

Lack of Transparency

Citizens lack real-time visibility into the status of their reports, breeding frustration and distrust in local governance.

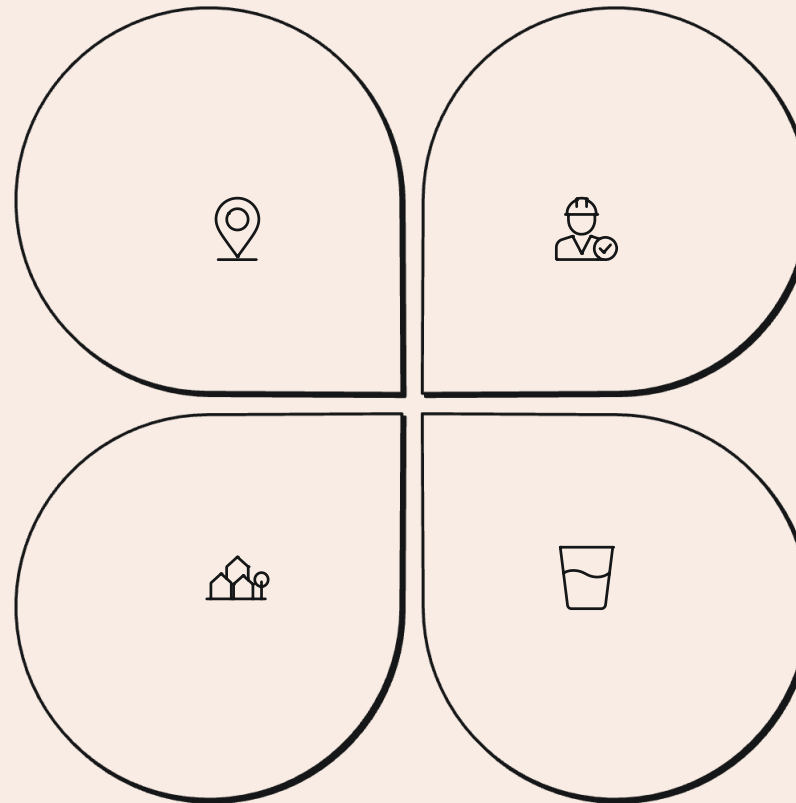
Our Objective: Bridging Citizens and Authorities

Instant Reporting

Enable citizens to report civic issues instantly using location and imagery.

Community Building

Encourage active citizen participation in the maintenance and improvement of their city.



Efficient Management

Provide administrators with tools to efficiently manage, verify, and resolve issues.

Accountability

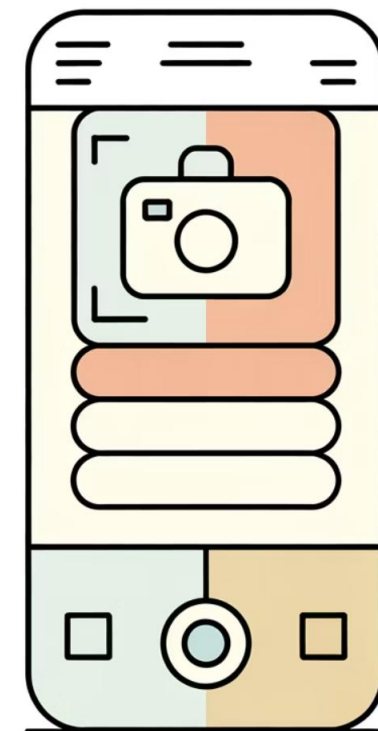
Introduce a transparent, trackable system to foster greater accountability in local services.

Introducing CivicConnect: Mobile Reporting Made Easy

A Seamless Reporting Workflow

- Users capture an image of the issue on their Android device.
- The app automatically detects and registers the precise geographical location (GPS).
- Users categorise the issue (e.g., Road, Water, Waste) and submit it.
- The report is instantaneously delivered to the Admin Panel for verification and action.
- Admins can then update the status, which is reflected in the user's dashboard.

Built entirely using Kotlin, CivicConnect ensures a native, responsive, and reliable experience for all users.



Key Features: Intelligence and Transparency



Location-Based Reporting

Automatic GPS detection ensures accurate placement of reported issues for rapid response.



AI-Generated Priority Score

Utilises the Gemini API to calculate urgency and automatically prioritise critical civic issues.



Real-Time Tracking

Citizens receive instant updates on their report status: Pending, In Progress, or Resolved.

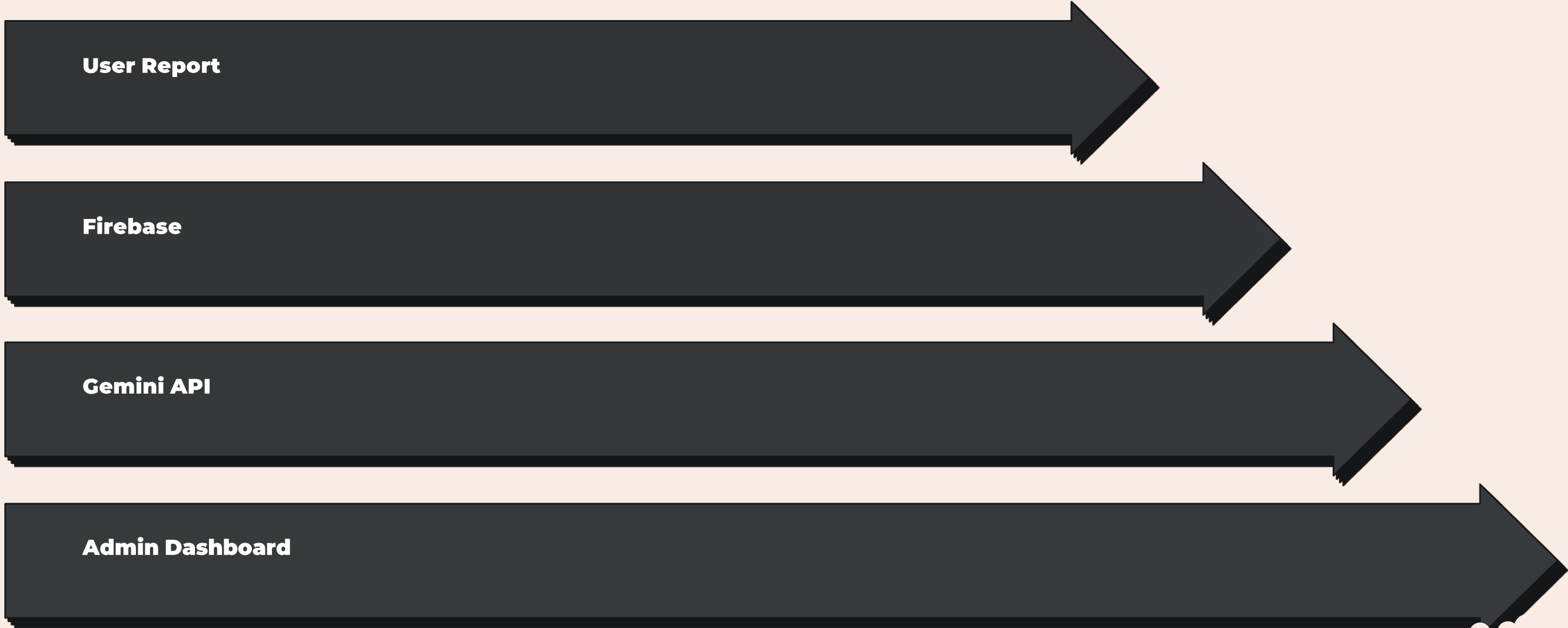


Secure & Scalable Backend

Leverages Firebase Authentication and Firestore for robust data management.

System Architecture: Data Flow

Our backend is designed for speed and reliability, ensuring smooth communication between all stakeholders, underpinned by Google's cloud infrastructure.



User Report

Firebase

Gemini API

Admin Dashboard

Technology Stack: Modern and Efficient



Mobile Frontend

Built natively with **Kotlin** and **XML** for performance and native Android look and feel.



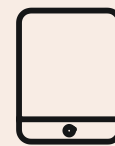
Cloud Backend

Uses **Firebase** (Auth, Firestore, Storage) for real-time data synchronisation and user management.



AI & Intelligence

Gemini API is integrated for advanced analytical tasks, specifically priority scoring of reports.



Design Pattern

Employing **MVVM Architecture** for clean separation of concerns and easier maintenance.

Impact: Benefits for the Citizen



- **Effortless Reporting**

Report issues in seconds with a photo and a tap, eliminating complicated official forms.

- **Informed Participation**

Stay continuously updated on the progress of their complaints via real-time notifications.

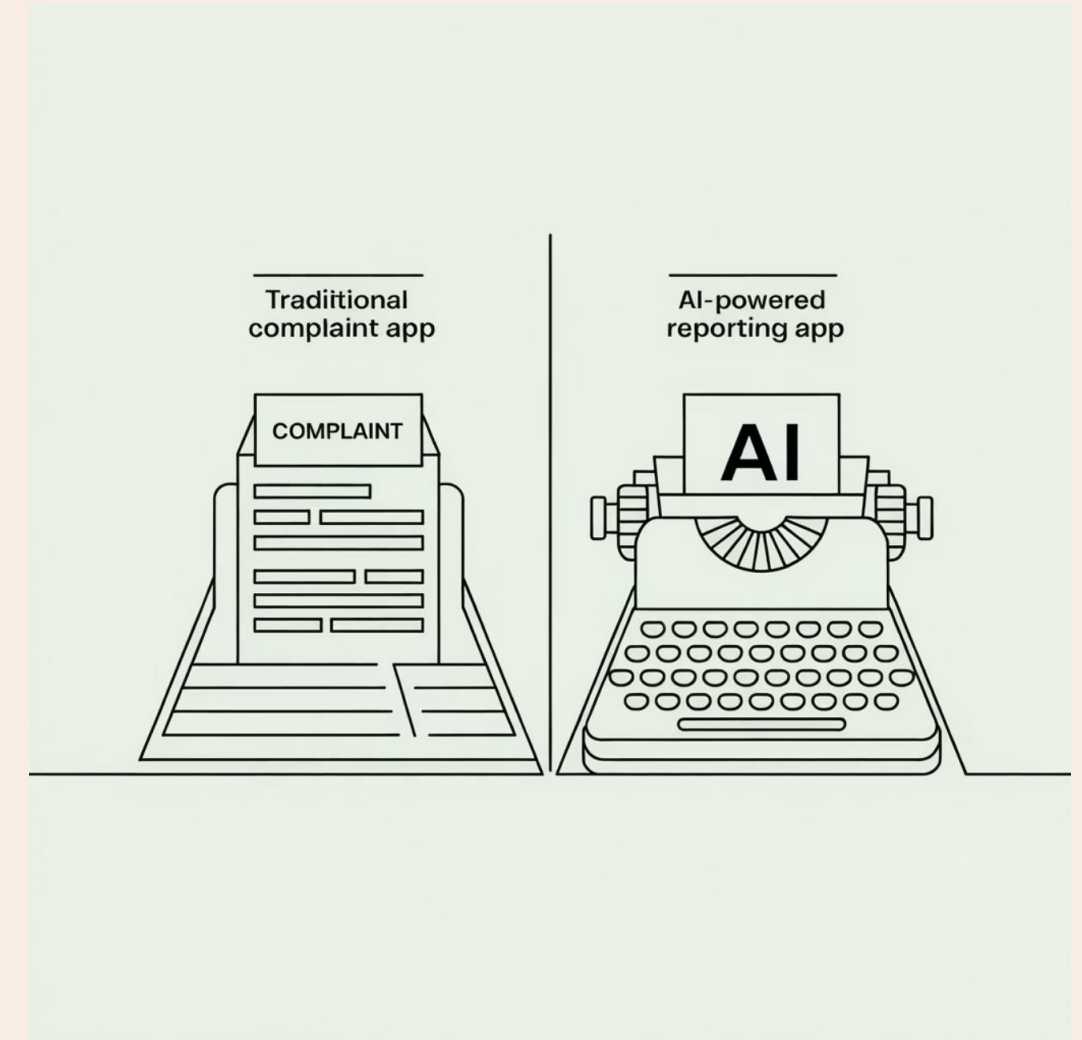
- **Empowered Community**

Fosters a sense of ownership and accountability, leading to more engaged community participation.

The CivicConnect Advantage: AI-Driven Prioritisation

While many existing apps simply log complaints, CivicConnect is differentiated by its focus on intelligent workflow and administrative efficiency.

- **AI Urgency Ranking:** Uses the Gemini API to assess the severity of the issue and assign a Priority Score, ensuring urgent problems are tackled first.
- **Dual Focus:** Designed for both **citizen convenience** and **administrative efficiency** in handling large volumes of reports.
- **Performance:** The Firebase backend guarantees lightning-fast data handling and reliability, crucial for public service applications.



Future Vision and Enhancements

To scale CivicConnect into a comprehensive city management tool, we plan the following additions:

	Advanced Analytics Implementing dashboards for tracking issue trends and identifying service hotspots.
	Administrative Integration Seamlessly connecting with local government officials on the same platform.
	AI Based Prioritization AI driven prioritization to ensure that the most critical issues are addressed first.
	Similarity Detection Using AI to group duplicate reports, reducing redundant administrative work.